VIKRAM EJJAGIRI

Applied AI Engineer | Industrial AI | Digital Production Management | Industry 4.0

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VIKRAM EJJAGIRI

Master's student in Applied AI for Digital Production Management at Technische Hochschule Deggendorf with a Bachelor's degree in Mechanical Engineering (Honors in Robotics). Experienced in developing AI-driven solutions for manufacturing, machine learning applications, industrial data analytics, predictive maintenance, and intelligent automation. Proficient in Python, Machine Learning, Power BI, SQL, and data-driven process optimization. Passionate about Industrial AI, smart manufacturing, digital production systems, and Industry 4.0.

PROJEKTE

Dashboard for Predictive Maintenance of Manufacturing Equipment
Objective: Development of a data-driven predictive maintenance solution for monitoring machine condition using temperature and vibration data to support intelligent maintenance planning.
Achievements: Processed and analyzed manufacturing sensor data using Python. Cleaned and prepared production datasets for machine learning applications. Developed and evaluated machine learning models to predict potential equipment failures. Built an interactive Power BI dashboard for monitoring machine health, maintenance KPIs, and production performance. Demonstrated how data-driven monitoring supports predictive maintenance and improves equipment availability.
Tools Used: Python, Pandas, Scikit-learn, SQL, Power BI, DAX, Excel

Intelligent Manufacturing Incident Assistant
Objective: Development of an AI-powered manufacturing incident analysis system for intelligent issue classification and operational decision support in smart factory environments.
Achievements: Developed an NLP-based incident analysis pipeline capable of automatically categorizing manufacturing incidents, assessing severity levels, and generating engineering recommendations using transformer-based models. Simulated AI-assisted production support workflows for Industry 4.0 manufacturing systems and implemented intelligent incident interpretation using GPU-accelerated inference pipelines.
Tools Used: Python, PyTorch, Hugging Face Transformers, NLP, Pandas

Generative Engineering Concept Studio
Objective: Development of an AI-powered engineering concept generation system for industrial product ideation and visualization using diffusion-based generative AI models.
Achievements: Developed a prompt-driven engineering concept generation pipeline capable of producing engineering-inspired design concepts using Stable Diffusion. Implemented GPU-accelerated inference workflows and automated image generation for rapid concept exploration. Demonstrated AI-assisted engineering ideation for product development and industrial design applications.
Tools Used: Python, PyTorch, Stable Diffusion, Hugging Face Diffusers, CUDA, Gradio, Prompt Engineering

Intelligent SCARA Robotic System for Multi-Plane Operations
Objective: Enhancement of a standard SCARA robot for multi-plane operations with 3D movement to increase industrial application flexibility.
Achievements: Redesigned the mechanical structure and positioning system for enhanced operational flexibility. Performed mechanical design validation and dynamic simulation to evaluate robot motion and positioning accuracy. Applied robotics, automation, and engineering principles to improve assembly operations across multiple planes.
Tools Used: Fusion 360, AutoCAD, MSC Adams, Python

EDUCATION

M.Eng. Applied AI for Digital Production Management, *Technische Hochschule Deggendorf* 03/2025 – Präsent
Relevant Coursework Deutschland

- Artificial Intelligence for Digital Production
- Machine Learning
- Data Analytics
- Smart Manufacturing
- Industry 4.0
- Digital Twin
- MES & ERP Concepts
- Robotics & Automation

- Cyber-Physical Production Systems

B.Eng. Mechanical Engineering (Honors in Robotics), *Vasavi College of Engineering* 08/2021 – 05/2024
- Bachelor's Thesis: Modified SCARA Robot Hyderabad, Indien
- Honors Thesis: Design and Analysis of Ornithopter
- CGPA: **8.17/10**

Diploma in Mechanical Engineering, *Government Polytechnic* 10/2007 – 09/2016
CGPA: **9.52/10** Warangal, India

TECHNICAL SKILLS

Programming — Python • SQL • Power BI • Excel

Artificial Intelligence & Machine Learning — Machine Learning • Deep Learning • Generative AI • NLP • Computer Vision • PyTorch • Scikit-learn

Data Analytics — Pandas • NumPy • Data Analysis • Data Visualization • Power Query • DAX

Industrial AI & Manufacturing — Predictive Maintenance • Smart Manufacturing • Industry 4.0 • Manufacturing Analytics • Process Optimization • Digital Production Systems

Robotics & Engineering — Robotics • Industrial Automation • SolidWorks • Fusion 360 • AutoCAD • MSC Adams • PLC (Basic) • ROS (Basic)

LANGUAGES

English
Professional Proficiency

German
B1

ACCOMPLISHMENTS

Overall State 43rd rank in Engineering Common Entrance Test/ ECET, *ECET* 01/11/2020

ACCOMPLISHMENTS

President of Robotics Club 01/12/2023 –
- Organized Line Follower & Maze Solver Robot workshops 01/12/2023

TRAININGS AND CERTIFICATES

- Advanced Robotics - NPTEL (2024)
- Design of HVAC System in Pharmaceutical - KG MECH (2022)
- Industrial Training on Lathe Machines - IRA OM Technologies Pvt. Ltd. (2021)